

Homework 3

Advanced Statistics and Data Analysis

Davide Risso

Sophie Grace Parolin

Wolff and Gorb ([2013](#)) studied the different frictional coefficients on the different legs of a spider. Please, read the article, with particular focus on the analysis described in its Figure 4. Briefly, the goal is to compare the pulling and pushing motions of different leg pairs. Load the data available in [this csv file](#).

Clear all objects from the environment to ensure a clean workspace

```
rm(list=ls())
```

Load required packages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readr)
```

Read the dataset

```
spider = read_csv("spider_wolff_gorb_2013.csv", skip=1)
```

Rows: 282 Columns: 3

-- Column specification -----

Delimiter: ","

chr (2): leg, type

dbl (1): friction

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Display the structure of the dataset

```
glimpse(spider)
```

Rows: 282

Columns: 3

\$ leg <chr> "L1", "L1", "L1", "L1", "L1", "L1", "L1", "L1", "L1", "L1", "~

\$ type <chr> "pull", "pull", "pull", "pull", "pull", "pull", "pull", "pull", "pull", "pull", ~

\$ friction <dbl> 0.90, 0.91, 0.86, 0.85, 0.80, 0.87, 0.92, 0.83, 0.88, 0.87, 0~

Convert leg and type variables to factors

```
spider <- spider |>
  mutate(leg = factor(leg), type = factor(type))
```

Perform the following analyses:

1. Visual inspection of the data: use a boxplot or similar plot to compare the distribution of the different forces (pull and push) and legs.

```
spider |>
  ggplot(aes(x = leg, y = friction, fill = type)) +
  geom_boxplot() +
  labs(title = "Friction Distribution by Force Type and Leg",
       x = "Leg",
       y = "Friction",
       fill = "Force Type") +
  theme_minimal()
```

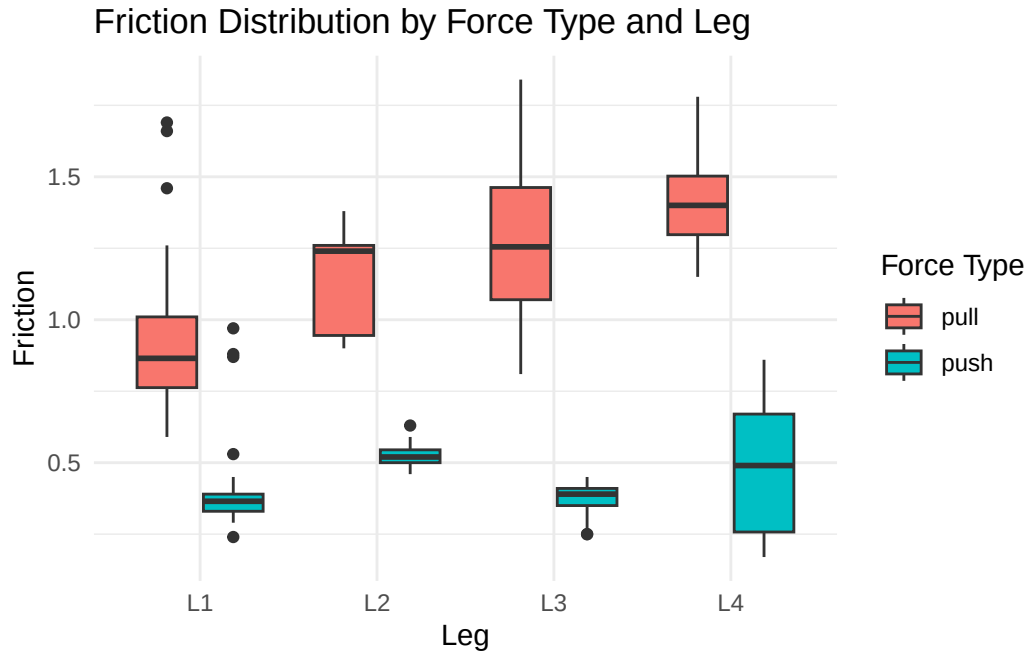


Figure 1

We can see that the pull force generally results in higher friction values compared to the push force across all legs. This suggests that pulling generates more resistance than pushing, possibly due to differences in mechanical interaction with the surface.

The front legs (L1, L2) and middle leg (L3) exhibit similar trends. While the fourth leg (L4) shows a different pattern. In fact, the friction values for push are more variable, with a wider interquartile range (IQR), indicating greater dispersion in the data. The increased variability in the push condition of L4 could indicate that this leg plays a unique role in locomotion.

2. Specify a linear model with friction as response and type as covariate. What is the effect of the force type on friction?

Fit the linear model

```
lm_type <- lm(friction ~ type, data = spider)
```

Display the summary of the model

```
summary(lm_type)
```

```

Call:
lm(formula = friction ~ type, data = spider)

Residuals:
    Min       1Q   Median       3Q      Max
-0.6212 -0.1290 -0.0222  0.1288  0.6288

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.21121    0.01946   62.23  <2e-16 ***
typepush    -0.77901    0.02752  -28.30  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2311 on 280 degrees of freedom
Multiple R-squared:  0.741, Adjusted R-squared:  0.7401
F-statistic: 801 on 1 and 280 DF, p-value: < 2.2e-16

```

The estimated linear model based on the regression output is:

$$\widehat{\text{friction}} = 1.21 - 0.78 \cdot \text{type}_{\text{push}}$$

The intercept (1.21) represents the estimated friction for the pull force (the reference category). The coefficient for `typepush` (-0.78) indicates that, on average, the friction is 0.78 units lower for the push force compared to the pull force. The p-value ($< 2e - 16$) suggests that this difference is highly statistically significant.

In conclusion, as expected from Figure 1, the force type significantly affects friction, with the push force generating significantly less friction than the pull force.

3. Specify a linear model with `friction` as response and `type` and `leg` as covariates. What is the effect of the force type on friction? Did it change compared to the previous model? What is the effect of `leg`?

Fit the linear model with both `type` and `leg` as covariates

```
lm_type_leg <- lm(friction ~ type + leg, data = spider)
```

Display the summary of the model

```
summary(lm_type_leg)
```

Call:

```
lm(formula = friction ~ type + leg, data = spider)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.46392	-0.13441	-0.00525	0.10547	0.69509

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.05392	0.02816	37.426	< 2e-16 ***
typepush	-0.77901	0.02482	-31.380	< 2e-16 ***
legL2	0.17192	0.04569	3.763	0.000205 ***
legL3	0.16049	0.03251	4.937	1.37e-06 ***
legL4	0.28134	0.03438	8.183	1.01e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2084 on 277 degrees of freedom

Multiple R-squared: 0.7916, Adjusted R-squared: 0.7886

F-statistic: 263 on 4 and 277 DF, p-value: < 2.2e-16

The estimated linear model based on the regression output is:

$$\widehat{\text{friction}} = 1.05 - 0.78 \cdot \text{type}_{\text{push}} + 0.17 \cdot \text{leg}_{L2} + 0.16 \cdot \text{leg}_{L3} + 0.28 \cdot \text{leg}_{L4}$$

```
table(spider$leg, spider$type)
```

	pull	push
L1	34	34
L2	15	15
L3	52	52
L4	40	40

The coefficient associated with `typepush` remains unchanged (-0.78) even after including the variable `leg` in the model, and the statistical significance ($p\text{-value} < 2e-16$) persists. This indicates that the effect of force type (`type`) on friction is independent of the leg used (`leg`). This

independence is guaranteed by design: as shown in the contingency table, each leg was tested exactly the same number of times with both force types (pull and push). The experiment therefore follows a balanced design, ensuring orthogonality between `type` and `leg`.

Regarding the effect of `leg` on `friction`, the reference category (baseline) for `leg` is L1, meaning that friction increases by 0.17 ($p < 0.001$) for L2, 0.16 ($p < 0.0001$) for L3, and 0.28 ($p < 0.0001$) for L4 compared to L1.

```
anova(lm_type, lm_type_leg)
```

Analysis of Variance Table

```
Model 1: friction ~ type
Model 2: friction ~ type + leg
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     280 14.956
2     277 12.035  3     2.9209 22.41 5.082e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA test confirms that `leg` significantly affects friction. Additionally, the inclusion of `leg` in the model improves its explanatory power, as reflected in the increase of the adjusted R^2 from 0.74 to 0.79.

4. Specify a contrast to compare leg pairs L3 and L2. Is the difference in friction significant?

```
library(contrast)
```

Perform a contrast to compare the estimated friction between legs L3 and L2 and display the result

```
contrast::contrast(lm_type_leg,
  list(type = "push", leg = "L3"),
  list(type = "push", leg = "L2"))
```

lm model parameter contrast

Contrast	S.E.	Lower	Upper	t	df	Pr(> t)
-0.01142949	0.04319685	-0.0964653	0.07360632	-0.26	277	0.7915

Since the difference in friction between L3 and L2 is small and non-significant ($p = 0.79$), we conclude that these two legs exhibit similar friction behavior. The results indicate that friction does not vary significantly between L3 and L2.

Note: The same results would be obtained using “pull” instead of “push” because `type` and `leg` are orthogonal by design. This orthogonality ensures that the estimated effect of `leg` is independent of the level of `type` used in the contrast.

5. Specify a linear model that includes, in addition to the two covariates, their interaction. Is the interaction significant? How do you interpret the coefficients?

Fit the linear model with interaction

```
lm_type_leg_interaction <- lm(friction ~ type * leg, data = spider)
summary(lm_type_leg_interaction)
```

Call:

```
lm(formula = friction ~ type * leg, data = spider)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.46385	-0.10735	-0.01111	0.07848	0.76853

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.92147	0.03266	28.215	< 2e-16 ***
typepush	-0.51412	0.04619	-11.131	< 2e-16 ***
legL2	0.22386	0.05903	3.792	0.000184 ***
legL3	0.35238	0.04200	8.390	2.62e-15 ***
legL4	0.47928	0.04442	10.789	< 2e-16 ***
typepush:legL2	-0.10388	0.08348	-1.244	0.214409
typepush:legL3	-0.38377	0.05940	-6.461	4.73e-10 ***
typepush:legL4	-0.39588	0.06282	-6.302	1.17e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1904 on 274 degrees of freedom

Multiple R-squared: 0.8279, Adjusted R-squared: 0.8235

F-statistic: 188.3 on 7 and 274 DF, p-value: < 2.2e-16

The fitted model is:

$$\hat{\text{friction}} = 0.92 - 0.51 \cdot \text{type}_{\text{push}} + 0.22 \cdot \text{legL}_2 + 0.35 \cdot \text{legL}_3 + 0.48 \cdot \text{legL}_4 - \\ 0.10 \cdot \text{type}_{\text{push}} : \text{legL}_2 - 0.38 \cdot \text{type}_{\text{push}} : \text{legL}_3 - 0.40 \cdot \text{type}_{\text{push}} : \text{legL}_4$$

This model includes interaction terms, indicating that the effect of **type** on **friction** depends on the level of **leg**. In other words, the relationship between force type and friction is not uniform across legs.

The intercept corresponds to the estimated friction for the **pull** force applied to leg L1, which serves as the reference category. The coefficient for **typepush** indicates that switching from **pull** to **push** reduces friction by 0.51 units, but this interpretation holds only for leg L1, due to the presence of interaction terms. Similarly, the coefficients for **legL2**, **legL3**, and **legL4** represent the differences in friction with respect to L1 under the **pull** condition: friction increases by 0.22 units for L2, 0.35 for L3, and 0.48 for L4.

The interaction terms quantify how the effect of the **push** force deviates from its effect on L1 for the other legs. Specifically, the interaction effects for L3 and L4 are substantial (−0.38 and −0.40, respectively), indicating that the reduction in friction when switching to **push** is even greater for these legs. For example, the total reduction in friction for L3 is $0.51 + 0.38 = 0.89$ units. Conversely, the interaction term for L2 is small and not statistically significant (−0.10), suggesting that the reduction in friction for **push** relative to **pull** is similar for L2 and L1.

This implies that the push force produces a greater reduction in friction for L3 and L4 than it does for L1 and L2. Thus, the effectiveness of the push force varies depending on the leg, justifying the inclusion of the interaction term in the model.

The boxplot in Figure 1 visually supports these findings. While the friction values under **push** are consistently lower than those under **pull**, the magnitude of the reduction differs across legs. In particular, the difference between **pull** and **push** is small for L1 and L2, whereas for L3 and L4, the drop is much more pronounced. This visual evidence reinforces the statistical conclusion that the effect of **type** on **friction** is not constant across legs.

Let's compare the model without and with interaction using ANOVA

```
anova(lm_type_leg, lm_type_leg_interaction)
```

Analysis of Variance Table

Model 1: `friction ~ type + leg`

Model 2: `friction ~ type * leg`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	277	12.0347				
2	274	9.9368	3	2.0979	19.282	2.256e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The ANOVA test comparing the models with and without interaction yields a highly significant result ($p < 0.001$), confirming that the interaction between **type** and **leg** significantly affects friction. This means that the effect of force type on friction depends on the leg used, justifying the inclusion of the interaction term in the model.

6. Specify a contrast to compare the push and pull force of leg L2. Is the difference significant?

```
contrast::contrast(  
  lm_type_leg_interaction,  
  list(type = "push", leg = "L2"),  
  list(type = "pull", leg = "L2")  
)
```

lm model parameter contrast

Contrast	S.E.	Lower	Upper	t	df	Pr(> t)
-0.618	0.0695372	-0.7548951	-0.4811049	-8.89	274	0

The contrast analysis confirms that, for leg L2, the push force results in a statistically significant reduction in friction compared to the pull force. The estimated difference in friction is -0.618 units, with a 95% confidence interval ranging from -0.75 to -0.48 , and a p-value effectively equal to zero.

This result is consistent with the model's coefficients: the estimated reduction for L2 combines the main effect of **typepush** (-0.51) and the interaction term for L2 (-0.10), totaling -0.61 , which aligns closely with the contrast estimate. Therefore, the overall effect of push is significantly lower than pull for this leg.